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Lab 4: Functional Requirement Elaboration using Use Case Scenario Template
Title of the Case Study: TourMate – An Intelligent Tourism Management Platform

1. Aim

The aim of this assignment is to analyze the TourMate Tourism Management System and understand how system requirements and development models are selected. The objectives are:

- Identify the initial system requirements using the MoSCoW prioritization technique.
- Compare different software development process models.
- Select the most suitable model for developing the TourMate system.

2. Problem Statement

The tourism industry involves multiple services such as tour planning, hotel reservations, transportation management, customer handling, and payment processing. Many tourism agencies still depend on manual systems or disconnected software tools, which often leads to errors, slow service, and poor coordination.

TourMate is proposed as a web-based Tourism Management System that helps automate these processes.The system allows customers to explore tour packages, make bookings, complete payments, and provide feedback. Administrators can manage tour packages, track bookings, and generate reports from a single platform.

3. Functional Requirements

3.1 Stakeholders

- End Users – Tourists: Individuals who use the system to explore and book tour packages.
- Administrator: Manages system operations, users, and tour packages.
- Tour Operators / Agencies: Provide and manage tour packages listed on the platform.
- Hotels and Transport Providers: Offer accommodation and transportation services integrated with the system.

3.2 Use Case: User Registration

Use Case Name: User Registration
Primary Actor: Customer / Tourist

- **Description:** This use case allows a new user to create an account in the TourMate Tourism Management System.
- **Preconditions:**
  - User must have access to the TourMate website.
  - User should not already have an account.
- **Main Flow:**
  - User opens the TourMate website.
  - User clicks on the register option.
  - User enters personal details (name, email, password, phone number).
  - User submits the registration form.
  - System validates the entered information.
  - System stores user data in the database.
  - System displays a successful registration message.
- **Alternative Flow:**If the email is already registered, the system shows an error message.
- **Postconditions:**A new user account is successfully created.

3.3 Use Case: User Login

Use Case Name: User Login
Primary Actor: Customer / Tourist

- **Description:**This use case allows registered users to log into the system.
- **Preconditions:** User must already be registered.
- **Main Flow:**
  - User opens the login page.
  - User enters registered email and password.

    - System verifies credentials.
    - System grants access to the dashboard.
- **Alternative Flow:**If credentials are incorrect, an error message is displayed.
- **Postconditions:** User successfully logs into the system.

3.4 Use Case: Browse Tour Packages

Use Case Name: View Tour Packages
Primary Actor: Customer / Tourist

- **Description:**Allows users to browse available tour packages.
- **Preconditions:**Tour packages must exist in the system.
- **Main Flow:**
    - User logs into the system.
    - User navigates to the Tour Packages section.
    - System displays available packages.
    - User selects a package to view details.
- **Postconditions:**User can view package details (destination, price, duration).

3.5 Use Case: Book Tour Package

Use Case Name: Book Tour Package
Primary Actor: Customer / Tourist

- **Description:**Allows users to book a selected tour package.
- **Preconditions:**
    - User must be logged in.
    - Tour packages must be available.
- **Main Flow:**
    - User selects a tour package.
    - System displays package details.
    - User enters booking details.
    - User confirms booking.
    - System redirects to payment page.
    - User completes payment.
    - System generates confirmation message.
- **Alternative Flow:**If payment fails, a failure message is displayed.
- **Postconditions:**Booking details are stored in the database.

3.6 Use Case: Admin Manage Tour Packages

Use Case Name: Manage Tour Packages
Primary Actor: Administrator

- **Description:**Allows admin to manage tour packages.
- **Preconditions:**Admin must be logged in.
- **Main Flow:**
    - Admin accesses dashboard.
    - Admin selects “Manage Tour Packages”.
    - Admin adds or updates packages.
    - Admin saves changes.
    - System updates database.
- **Postconditions:**Tour package data is updated successfully.

Conclusion

The functional requirements of the TourMate Tourism Management System outline some of the essential operations that need to take place within the system to ensure proper support for users. The definition of these requirements based on use case scenarios such as user registration, login, browsing of tour packages, booking of tours, and management of packages allows an understanding of how users interact with the system. This is essential in ensuring that all functionalities are implemented appropriately. The functional requirements offer a clear foundation for designing and developing the TourMate system efficiently.